

where γ is set to three. Funds can be ranked on this measure from high to low, with the best funds having the highest values. Equation (10.16) harks back to chapter 2 on utility functions, and indeed it is a CRRA or power utility function. (More precisely, it's the certainty equivalent of a CRRA utility function.) Goetzmann et al. report that Morningstar uses a variant of this measure:

$$\left(\frac{1}{T} \sum_{t=1}^T \frac{1}{(1 + r_t - r_{ft})^2} \right)^{-\frac{1}{2}} - 1,$$

which is also a CRRA utility function with risk aversion $\gamma = 2$.

3.5 DOES ALPHA EVEN EXIST?

Since alpha is based on a benchmark and estimates of alpha are very sensitive to that benchmark, is there even such a thing as true alpha? It could just be a wrong benchmark. The academic literature calls this a *joint hypothesis problem*, and the search for alpha is the same as the testing for market efficiency.²¹ In a major contribution, Hansen and Jagannathan (1997) show that it is always possible to find an ex-post benchmark portfolio that produces no alpha. This is less useful ex ante, but it shows that a benchmark portfolio can always be constructed where no alpha exists after the fact. Since Grossman and Stiglitz (1980), the profession recognizes that perfectly efficient markets cannot exist (see chapter 6)—so there is alpha—but as the analysis of this section has shown, even for a recognized master of investing like Buffett, alpha can be very hard to detect statistically.

The joint hypothesis problem—that alpha and the benchmark are simultaneously determined—is the key problem for asset owners. It is of little use for an academic to say that Fidelity has no alpha, when the asset owner cannot access the complicated size, value-growth, and momentum factors used to compute that alpha. For that asset owner, Fidelity may be providing alpha. For another asset owner, Fidelity may well be adding negative alpha because she can do all the appropriate factor exposure (and implement the underlying replicating factor benchmark portfolios) on her own.

Choosing the right set of factors, then, is the most relevant issue for asset owners, and I return to this topic in chapter 14. Alpha is primarily a statement about the factor benchmark (or lack of a factor benchmark). We now have enough knowledge of risk adjustments to judge different alpha opportunities, and so we turn to one source of alpha that has recently stirred up debate.

²¹For a summary of this large literature, see Ang, Goetzmann, and Schaefer (2011).